AI 101 for Mechanical Engineers · Module 3

Available inputs and causal claims

Lesson 3.3

What you will learn

> Check whether an input exists at decision time, and distinguish statistical association, predictive performance and causal claims.

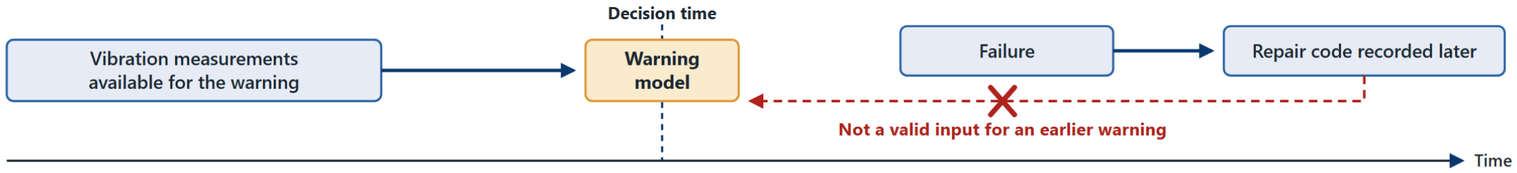
Lesson overview

Module 3 · Lesson 3

Available inputs and causal claims

AI 101 for Mechanical Engineers

1 Use only information available at decision time



Using later information is **information leakage**. A signal window must also exclude readings taken after the decision.

2 Correlation

Do the variables vary together?

Flow rate and motor current increase together in the records.

Flow rate changes | Motor current changes

An association, not proof of a cause.

3 Prediction

Can available inputs estimate outputs on relevant unseen examples?

Available inputs → Model → Estimated output

Check performance under the conditions that matter.

Accurate prediction alone does not prove causation.

4 Causation

Does changing something change the outcome?

Test how a controlled change in flow affects motor current.

Interpret the result using pump physics, controls and test conditions.

A causal claim needs physical and experimental evidence.

Key takeaway

Match the claim to the evidence.

A sensor that rises with a fault is not a control: changing its recorded value does not repair equipment.

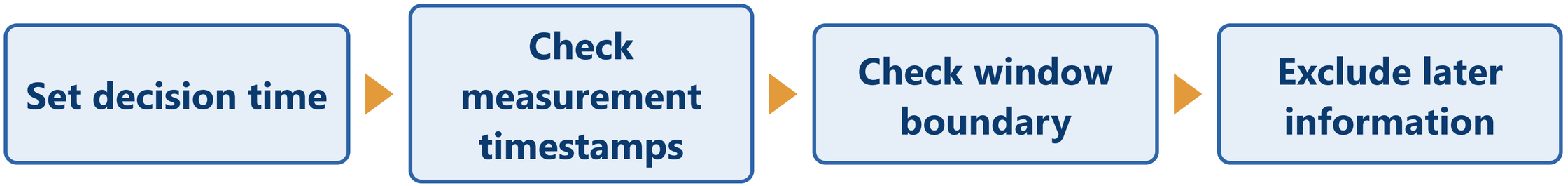
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Use only inputs available at decision time

- A valid predictor uses information available at the decision
- A post-failure repair code cannot predict that failure beforehand
- It gives information absent when the warning was needed
- This is information leakage [25]

Check every input against the decision time



- A signal-window feature can also leak
- Leaks if the window includes measurements after the decision

Three different questions

Claim	Question	Evidence
Correlation	Do variables vary together?	Observational records
Prediction	Do inputs estimate outputs on unseen examples?	Evaluation on unseen data
Causation	Does an intervention change the outcome?	Controlled intervention evidence

> Successful prediction shows useful estimation under evaluated conditions, not that an input causes the output.

Association is not a causal relationship

Records show

- Flow rate and motor current increase together
- This is an association

A causal claim needs

- Evidence about interventions [26]
- Pump physics, controls, experimental conditions

A predictive coefficient alone does not establish the causal relationship.

Changing a sensor reading repairs nothing

- Load can change both a temperature reading and fault
- So the sensor can correlate with the fault
- Changing the recorded value would not repair the equipment
- Investigate the physical system before recommending an intervention

CHECK YOUR UNDERSTANDING

Which input would be invalid for a warning issued before a failure: current vibration information or a repair code recorded after the failure?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Which input would be invalid for a warning issued before a failure: current vibration information or a repair code recorded after the failure?

> The post-failure repair code: it was unavailable when the warning had to be issued. A vibration feature must also exclude measurements after decision time.

CHECK YOUR UNDERSTANDING

Does a predictive association establish physical causation?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Does a predictive association establish physical causation?

> No. A causal claim needs evidence about interventions under appropriate physical and experimental conditions.

Key takeaways

- Verify input availability at the intended decision time
- Exclude post-event information and future measurements
- Correlation, prediction and causation answer different questions
- Use physical and experimental evidence for causal interventions

References

[25] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, “Leakage in data mining: Formulation, detection, and avoidance,” ACM Trans. Knowl. Discov. Data, vol. 6, no. 4, pp. 1–21, 2012. doi: 10.1145/2382577.2382579

[26] J. Pearl, Causality: Models, Reasoning, and Inference, 2nd ed. Cambridge, U.K.: Cambridge Univ. Press, 2009. doi: 10.1017/CBO9780511803161

Numbers match the course reference list. The written lesson is the authoritative text.